

LOW-STYRENE SBR PATENT LANDSCAPE: SOLUTION, EMULSION, FUNCTIONALIZATION, AND BINDER-FORMULATION ROUTES

1. ABSTRACT

This report synthesizes nine primary patents relevant to low-styrene styrene-butadiene rubber (SBR) synthesis, modification, and downstream use. The target is a styrene-butadiene elastomer biased toward butadiene, typically with reduced styrene fraction relative to conventional SBR grades, while preserving processability and performance. Across the selected patents, the dominant technical themes are solution anionic copolymerization in hydrocarbon solvent, emulsion/latex polymerization in water, post-polymerization functionalization or grafting, and formulation strategies that improve handling or end-use properties without increasing styrene content. Several patents explicitly disclose styrene-butadiene copolymers with styrene contents in the low-to-moderate range, including 3 wt%, 20.2 wt%, 21.3 wt%, 21.7 wt%, 22.1 wt%, 25 wt%, 25.7 wt%, and 27.0 wt%, depending on embodiment and application. The landscape also includes patents where SBR is not the central synthetic target but is used as a benchmark binder or rubber component in a broader formulation context.

The major polymerization approaches are: (1) solution anionic polymerization in nonpolar hydrocarbon solvents such as cyclohexane, hexane, methylcyclohexane, or n-hexane, often with lithium alkyl initiators, randomizers/polar additives, and post-polymerization coupling or branching; (2) aqueous emulsion polymerization for cross-linked SBR microsphere binders, using deionized water, emulsifiers, oxidizing/reducing systems, and extensive demulsification/washing workup; and (3) grafting or functionalization of preformed SBR to improve filler interaction, processability, or battery-binder performance. Process trends include precise control of monomer feed ratio, conversion, temperature, and residence time; use of chain-end modification or branching agents to tune macrostructure; and downstream water-based coagulation, stripping, washing, or extraction steps. Overall, the selected patents show that low-styrene SBR development is driven less by a single composition window than by integrated control of polymer microstructure, branching, and post-treatment to achieve tire or binder performance with reduced styrene fraction.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US6858683B2
- Patent Title: Process for the preparation of SBR rubbers with an improved processability and ...
- Assignee: Polimeri Europa S.P.A.
- Jurisdiction: US
- Publication Year: 2005
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution anionic copolymerization of butadiene and styrene in hydrocarbon solvent, using lithium alkyl initiators and randomizing agents, followed by post-polymerization branching/coupling chemistry. The examples compare linear, radial, and branched SBR architectures and then remove solvent by steam stripping and drying after oil extension. - Cyclohexane: 15 liters (11.7 Kg) — Comparative Example 1 - Butadiene: 1.14 Kg — Comparative Example 1 - Styrene: 0.38 Kg — Comparative Example 1 - Temperature: 30° C. — Comparative Example 1 - Temperature: 80° C. — Comparative Example 1 - Li n-butyl: 2.05 grams of a solution at 5% by weight — Comparative Example 1 - THFA: 200 ppm — Comparative Example 1 - Water stripping temperature: 100° C. — Comparative Example 1 - Vacuum oven drying temperature: 60° C. — Comparative Example 1 - Residual water content: less than 0.3% by weight — Comparative Example 1 - Residence time: 30 minutes — Comparative Example 2 - Butadiene: 9% by weight — Comparative Example 2 - Styrene: 3% by weight — Comparative Example 2 - Randomizing agent/vinyl promoter: 250 ppm — Comparative Example 2 - Lithium/randomizing agent ratio: 0.3-0.25 — Comparative Example 2 - Temperature: 57° C. — Comparative Example 2 - Temperature: 66° C. — Comparative Example 2 - Overall conversion: 99% — Comparative Example 2 - Butadiene: 1.14 Kg — Comparative Example 3 - Styrene: 0.38 Kg — Comparative Example 3 - THFA: 400 ppm — Comparative Example 3 - Temperature: 30° C. — Comparative Example 3 - Temperature: 80° C. — Comparative Example 3 - SiCl₄: 1.59 grams of a solution at 5% in cyclohexane — Comparative Example 3 - Vacuum oven drying temperature: 60° C. — Comparative Example 3 - Butadiene: 9% by weight — Comparative Example 4 - Styrene: 3% by weight — Comparative Example 4 - Randomizing agent/vinyl promoter: 350 ppm — Comparative Example 4 - Lithium/randomizing agent ratio: 0.5-0.6 — Comparative Example 4 - Temperature: 53° C. — Comparative Example 4 - Temperature: 61° C. — Comparative Example 4 - Temperature: 66° C. — Comparative Example 4 - SiCl₄: 40% of the lithium fed (equivalents/equivalents) — Comparative Example 4 - DVB ratio to first reactor vs second reactor: 1.4-1.5 — Comparative Example 9 - Li/DVB ratio to first reactor: 6.0 to 3.0 — Comparative Example 9 - Temperature: 55-57° C. — Comparative Example 9 - Temperature: 64-66° C. — Comparative Example 9 - Temperature: 67° C. — Comparative Example 9 - DVB total quantity: equal to the lithium fed — Comparative Example 10 - DVB distribution: 75% to first reactor and 25% to second reactor — Comparative Example 11 - Styrene content: always 25% by weight — Examples section - Vinyl units: 65-67% — Examples section -

Oil content: 27.3% by weight — Examples section - THF solvent: 25° C. — Examples section - Medium & Water Role: The polymerization medium is organic solution (cyclohexane/hydrocarbon solvent), not aqueous. Water appears later only in downstream solvent-removal/workup steps, including steam stripping and recovery of damp rubber. - low styrene content: 3% by weight

Medium & Water Role - The polymerization medium is organic solution (cyclohexane/hydrocarbon solvent), not aqueous. Water appears later only in downstream solvent-removal/workup steps, including steam stripping and recovery of damp rubber. - Core reaction medium: Organic hydrocarbon solution polymerization (cyclohexane / hydrocarbon solvent); later steam stripping and water-containing workup in some examples

low styrene content - 3% by weight

Relevant Experimental Evidence - Comparative Example 1: Linear batch SBR was prepared in cyclohexane and isolated by steam stripping and drying. - Comparative Example 2: Continuous linear SBR was produced in two CSTRs with 9 wt% butadiene and 3 wt% styrene feed. - Comparative Example 3: Radial batch SBR was obtained using SiCl₄ coupling after solution polymerization. - Comparative Example 4: Radial continuous SBR was produced in a three-reactor train with SiCl₄ coupling. - Example 5: Branched batch SBR was formed by adding bromine octyl and lithium n-butyl after polymerization. - Example 6: Branched continuous SBR was produced using on-line addition of Li n-butyl and Br-octyl after complete conversion.

Technical Relevance This patent is directly relevant because it is centered on SBR synthesis and explicitly includes low-styrene feed embodiments in solution polymerization. It also shows how branching and reactor design can be used to tune processability while maintaining a butadiene-rich composition.

Patent 2

- Patent Number: US12129321B2
- Patent Title: Cross-linked SBR microsphere binder and preparation method thereof and lithium- ...
- Assignee: Zhuhai Cosmx Battery Co., Ltd.
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an aqueous emulsion route to prepare a cross-linked SBR microsphere binder. Butadiene and styrene are pretreated with sodium hydroxide solution and deionized water, then polymerized in a deionized-water-based emulsion with emulsifier/co-emulsifier, redox components, and peroxide initiator, followed by latex modification, salt coagulation, acid treatment, extraction, and repeated

washing. - Sodium hydroxide aqueous solution: 8% mass fraction — Example 1, step 1 - Temperature: 20° C. — Example 1, step 1 - Time: 4 h — Example 1, step 1 - Deionized water: 200 parts — Example 1, step 2 - Styrene: 25 parts — Example 1, step 2 - n-heptane: 80 parts — Example 1, step 2 - n-propanol: 20 parts — Example 1, step 2 - methanol: 20 parts — Example 1, step 2 - glycerin: 40 parts — Example 1, step 2 - xylene: 40 parts — Example 1, step 2 - butadiene: 75 parts — Example 1, step 2 - dodecyl mercaptan: 0.16 parts — Example 1, step 2 - disproportionated rosin acid soap: 4.62 parts — Example 1, step 2 - sodium methylene bisnaphthalene sulfonate: 0.15 parts — Example 1, step 2 - sodium dithionite dihydrate: 0.025 parts — Example 1, step 2 - potassium phosphate: 0.24 parts — Example 1, step 2 - Temperature: 25° C. — Example 1, step 2 - ferrous sulfate: 0.01 parts — Example 1, step 2 - ethylenediamine tetraacetic acid-disodium salt: 0.01 parts — Example 1, step 2 - Temperature: 5° C. — Example 1, step 2 - cumene hydrogen peroxide: 0.06 parts — Example 1, step 3 - Pressure: 0.1 MPa — Example 1, step 3 - Polymerization temperature: 8° C. — Example 1, step 3 - Stirring speed: 100 r/min — Example 1, step 3 - Polymerization time: 7 h — Example 1, step 3 - sodium dimethyl dithiocarbamate: 0.1 parts — Example 1, step 3 - sodium nitrite: 0.02 parts — Example 1, step 3 - sodium polysulfide: 0.02 parts — Example 1, step 3 - polyvinylamine: 0.02 parts — Example 1, step 3 - sodium chloride solution: 20% — Example 1, step 4 - dilute sulfuric acid: 0.5% — Example 1, step 4 - Washing temperature: 50° C. — Example 1, step 4 - Extraction temperature: 50° C. — Example 1, step 4 - Particle size: 10 nm to 100 nm — Example 1 - Porosity: 0.01% to 3% — Example 1 - Pore diameter: greater than 0 nm and less than or equal to 10 nm — Example 1 - Sodium hydroxide aqueous solution: 10% mass fraction — Example 2, step 1 - Temperature: 40° C. — Example 2, step 1 - Time: 10 min — Example 2, step 1 - Deionized water: 400 parts total amount — Example 2, step 2 - Styrene: 50 parts — Example 2, step 2 - Butadiene: 150 parts — Example 2, step 2 - cumene hydrogen peroxide: 0.24 parts — Example 2, step 2 - Sodium chloride solution: 24% — Example 3, step 4 - Polymerization pressure: 0.25 MPa — Example 3, step 3 - Polymerization temperature: 7° C. — Example 3, step 3 - Polymerization time: 9 h — Example 3, step 3 - Particle size: 60 nm to 200 nm — Example 3 - Porosity: 8% to 12% — Example 3 - Sodium hydroxide aqueous solution: 14% mass fraction — Example 4, step 1 - Pressure: 0.3 MPa — Example 4, step 3 - Polymerization temperature: 5° C. — Example 4, step 3 - Polymerization time: 12 h — Example 4, step 3 - Sodium chloride solution: 30% — Example 4, step 4 - Extraction temperature: 70° C. — Example 4, step 4 - Particle size: 80 nm to 300 nm — Example 4 - Porosity: 12% to 20% — Example 4 - Sodium hydroxide aqueous solution: 16% mass fraction — Example 5, step 1 - Pressure: 0.15 MPa — Example 5, step 3 - Polymerization temperature: 7° C. — Example 5, step 3 - Polymerization time: 8 h — Example 5, step 3 - Sodium chloride solution: 26% — Example 5, step 4 - Particle size: 90 nm to 400 nm — Example 5 - Porosity: 29% to 35% — Example 5 - Sodium hydroxide aqueous solution: 9% mass fraction — Example 6, step 1 - Pressure: 0.25 MPa — Example 6, step 3 - Polymerization temperature: 6° C. — Example 6, step 3 - Polymerization time: 10 h — Example 6, step 3 - Sodium chloride solution: 25% — Example 6, step 4 - Sodium hydroxide aqueous solution: 8% mass fraction — Example 7, step 1 - Polymerization temperature: 8° C. — Example 7, step 3 - Polymerization time: 9 h — Example 7, step 3 - Sodium chloride solution: 28% — Example 7, step 4 - Particle size: 100 nm to 1 µm — Example 7 - Porosity: 20% to 29% — Example 7 - Medium & Water Role: Water is part of the core polymerization medium as a deionized-water-based emulsion/latex system. Water is also used later for coagulation, washing, and extraction workup of the latex-derived binder. - low

styrene content: 25 parts styrene

Medium & Water Role - Water is part of the core polymerization medium as a deionized-water-based emulsion/latex system. Water is also used later for coagulation, washing, and extraction workup of the latex-derived binder. - Core reaction medium: Aqueous emulsion / latex polymerization

low styrene content - 25 parts styrene

Relevant Experimental Evidence - Example 1: A porous cross-linked SBR microsphere binder was prepared in aqueous emulsion and isolated by salt coagulation, acid treatment, extraction, and washing. - Example 2: A broader formulation window was disclosed with deionized water, multiple solvents, and a modified monomer package. - Example 3: A higher-water, higher-styrene formulation produced microspheres with larger particle size and higher porosity. - Example 4: Increased solids and altered solvent package yielded a binder with 80-300 nm particles and 12-20% porosity. - Example 5: Another formulation variant produced 90-400 nm particles and 29-35% porosity.

Technical Relevance This patent is highly relevant because it directly prepares an SBR-based binder from styrene and butadiene in a waterborne emulsion system. Although not a low-styrene optimization patent per se, it discloses SBR compositions with styrene levels in the low-to-moderate range and a process architecture useful for low-styrene binder design.

Patent 3

- Patent Number: US7612144B2
- Patent Title: Use of sulfur containing initiators for anionic polymerization of monomers
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses anionic solution polymerization of styrene-butadiene copolymers using sulfur-containing initiators such as 2-lithio-2-methyl-1,3-dithiane and 2-lithio-2-phenyl-1,3-dithiane. The examples show polymerization in hexane or THF/hexane mixtures, followed by coagulation in isopropanol containing BHT and drum drying. - Tetrahydrofuran: 350 mL — Example No. 1 - 2-methyl-1,3-dithiane: 10 mL (83.5 mmol) — Example No. 1 - Butyllithium: 55.83 mL of 1.510 M (84.3 mmol) — Example No. 1 - Temperature: -78° C. — Example No. 1 - Temperature: -25° C. — Example No. 1 - 1-bromo-3-chloropropane: 8.26 mL (83.5 mmol) — Example No. 1 - Hexane: 1.12 kg — Example No. 2 - Styrene solution: 0.48 kg of 33 wt % styrene in hexane — Example No. 2 - Butadiene solution: 2.89 kg of 22.0 wt % butadiene in hexane — Example No. 2 -

Temperature: 24° C. — Example No. 2 - Cyclic oligomeric oxolanyl alkane modifier: 0.5 mL of 1.6 M — Example No. 2 - 2-lithio-2-methyl-1,3-dithiane: 22.63 mL of 0.234 M — Example No. 2 - Temperature: 54° C. — Example No. 2 - Reaction time: 15 minutes — Example No. 2 - Reaction time: 25 minutes — Example No. 2 - Hexane: 1.12 kg — Example No. 3 - Styrene solution: 0.48 kg of 33 wt % styrene in hexane — Example No. 3 - Butadiene solution: 2.89 kg of 22.0 wt % butadiene in hexane — Example No. 3 - Temperature: 24° C. — Example No. 3 - Cyclic oligomeric oxolanyl alkane modifier: 0.5 mL of 1.6 M — Example No. 3 - 2-lithio-2-methyl-1,3-dithiane: 16.96 mL of 0.234 M — Example No. 3 - Temperature: 54° C. — Example No. 3 - Reaction time: 17 minutes — Example No. 3 - Reaction time: 10 minutes — Example No. 3 - Hexane: 1.07 kg — Example No. 4 - Styrene solution: 0.48 kg of 33 wt % styrene in hexane — Example No. 4 - Butadiene solution: 2.95 kg of 21.6 wt % butadiene in hexane — Example No. 4 - Temperature: 24° C. — Example No. 4 - 2-methyl-1,3-dithiane: 8.47 mL of 0.5 M in hexane — Example No. 4 - Butyllithium: 3.42 mL of 1.55 M in hexanes — Example No. 4 - Temperature: 54° C. — Example No. 4 - Reaction time: 28 minutes — Example No. 4 - Temperature: 68.6° C. — Example No. 4 - Tetrahydrofuran: 5 mL — Example No. 14 - 2-phenyl-1,3-dithiane: 2.1 g (10.69 mmol) — Example No. 13 - Butyllithium: 6.37 mL of 1.68 M in hexane — Example No. 13 - Temperature: -78° C. — Example No. 13 - Temperature: 0° C. — Example No. 13 - 2-phenyl-1,3-oxathiane: 1.0 g (5.5 mmol) — Example No. 15 - Butyllithium: 3.3 mL of 1.68 M in hexane — Example No. 15 - Temperature: -78° C. — Example No. 15 - Temperature: -5° C. — Example No. 15 - Hexane: 1.610 kg — Example No. 35 - Styrene solution: 0.412 kg of 33 weight % styrene in hexane — Example No. 35 - Butadiene solution: 2.419 kg of 22.5 weight % butadiene in hexane — Example No. 35 - Temperature: 24° C. — Example No. 35 - Temperature: 50° C. — Example No. 35 - Reaction time: 5 to 10 minutes — Example No. 35 - Reaction time: 16 minutes — Example No. 35 - Reaction time: 15 minutes — Example No. 35 - DMI: 9.14 M — Example No. 35 - Temperature: 50° C. bath — Example No. 35 - Temperature: 1.68 M — Example No. 36 (initiator concentration) - Temperature: 50° C. — Example No. 36 - Temperature: 100° C. — Example No. 36 - Temperature: 50° C. — Example No. 45 - Temperature: 48.9° C. — Example No. 45 - Temperature: 171° C. — Example No. 12 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - low styrene content: 21.7% styrene

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent solution polymerization (hexane / THF / hydrocarbon solvent)

low styrene content - 21.7% styrene

Relevant Experimental Evidence - Example No. 2: A styrene-butadiene copolymer was prepared with 2-lithio-2-methyl-1,3-dithiane and showed 21.7% styrene. - Example No. 3: A similar copolymer gave 22.2% styrene and 31.2% vinyl content. - Example No. 4: In situ initiator formation was used to make a styrene-butadiene copolymer with 56.6° C. Tg. - Example No. 35: In-situ dithiane initiation and DMI termination produced functional SBR variants. - Example No. 45: A 2-lithio-2-methyl-1,3-dithiane initiated polymer had 20.2% styrene.

Technical Relevance This patent is relevant because it directly discloses solution-polymerized styrene-butadiene copolymers and their functionalized variants. The styrene contents are in the low-to-moderate SBR range, and the sulfur-containing initiators provide a route to tailor chain-end functionality without changing the basic SBR identity.

Patent 4

- Patent Number: EP3030430B1
- Patent Title: Modified diene elastomer comprising a diene elastomer coupled by an ...
- Assignee: Compagnie Générale des Etablissements Michelin
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution-polymerized diene elastomers modified by coupling with an alkoxysilane/amine-containing agent and then stabilized by condensation with a modification stabilizer. The examples prepare SBR in methylcyclohexane with n-butyllithium, then add methanol, antioxidant, and finally remove solvent by steam stripping and roll drying. - Nitrogen pressure: about 2 bars — Preparation of polymer A - Methylcyclohexane: 44.7 kg — Preparation of polymer A - Styrene: 1.8 kg — Preparation of polymer A - Butadiene: 4.9 kg — Preparation of polymer A - Tetrahydrofurfuryl ether solution: 395 mL of 0.1 mol.L⁻¹ in methylcyclohexane — Preparation of polymer A - n-butyllithium: 535 mL of 0.05 mol.L⁻¹ in methylcyclohexane — Preparation of polymer A - Temperature: 50 °C — Preparation of polymer A - Time: 40 minutes — Preparation of polymer A - Conversion rate: 90 % — Preparation of polymer A - Methanol solution: 530 mL of 0.15 mol.L⁻¹ in toluene — Preparation of polymer A - 4,4'-methylene-bis-2,6-tert-butylphenol: 0.8 phr — Preparation of polymer A - N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine: 0.2 phr — Preparation of polymer A - Steam stripping temperature: 100 °C — Preparation of polymer A - Roll drying temperature: 100 °C — Preparation of polymer A - Roll drying time: 15 minutes — Preparation of polymer A - Mooney viscosity: 60 — Preparation of polymer A - Mn: 192 000 g.mol⁻¹ — Preparation of polymer A - Ip: 1.07 — Preparation of polymer A - 1,2 units: 59 % relative to butadiene units — Preparation of polymer A - Styrene mass content: 25 % — Preparation of polymer A - Glass transition temperature: -24 °C — Preparation of polymer A - Medium & Water Role: The polymerization itself is carried out in organic solvent (methylcyclohexane). Water appears later only in steam stripping used to remove solvent before roll drying. - low styrene content: 25 %

Medium & Water Role - The polymerization itself is carried out in organic solvent (methylcyclohexane). Water appears later only in steam stripping used to remove solvent before roll drying. - Core reaction medium: Organic solution polymerization in methylcyclohexane; downstream steam stripping and drying

low styrene content - 25 %

Relevant Experimental Evidence - Preparation of polymer A: Non-functional SBR control was made in methylcyclohexane and steam stripped. - Preparation of polymer B: Functional amine SBR at chain end was prepared under the same base polymerization conditions. - The examples demonstrate that the SBR microstructure can be held at 25 wt% styrene while varying end functionality. - Steam stripping and roll drying were used after modification and stabilization.

Technical Relevance This patent is relevant because it explicitly discloses solution SBR with 25 wt% styrene and a controlled microstructure, which sits at the upper edge of the low-styrene range used in many tire grades. It also shows how post-polymerization stabilization can preserve processability after solvent removal.

Patent 5

- Patent Number: US7312264B2
- Patent Title: Process for obtaining a grafted elastomer having functional groups along the ...
- Assignee: Michelin Recherche Et Technique S.A.
- Jurisdiction: US
- Publication Year: 2007
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses radical grafting of mercaptan-type functional groups onto solution SBR, with a prior antioxidant treatment to suppress macrostructure change. The SBR is dissolved in cyclohexane, grafted at 80° C. with lauroyl peroxide and mercaptopropionic or mercaptoundecanoic acid, then stripped, coagulated if needed, and dried. - SBR mass: 45 g — Example 1 - Cyclohexane: 404 ml (315 g) — Example 1 - Temperature: 80° C. — Example 1 - Nitrogen bubbling: 10 minutes — Example 1 - Lauroyl peroxide / mercaptan molar ratio: 48 — Example 1 - Reaction time: 2 hours — Example 1 - 6-PPD: 0.4 phr — Example 1 - 6-PPD solution: 40 g/l — Example 1 - Hydrochloric acid: 37%-strength — Example 1 - pH: 2 — Example 1 - Drying temperature: 50° C. — Example 1 - Reduced pressure drying: yes — Example 1 - Water/acetone/HCl mixture proportions: 1/4/1 by volume — Example 1 - Starting SBR styrene units: 25.7% — Example 1 - Starting SBR 1,2 linkages: 43.0% — Example 1 - Starting SBR Tg: -37° C. — Example 1 - Starting SBR Mooney viscosity: 33.6 — Example 1 - 6-PPD: 0.39 phr — Example 2 - Mercaptopropionic acid: 150 meq./kg — Example 2 - Lauroyl peroxide molar ratio: 48 — Example 2 - 6-PPD: 0.20 phr — Example 2 - Mercaptoundecanoic acid: 150 meq./kg — Example 2 - Grafting yield: 58% — Example 2 - Grafting yield: 66% — Example 2 - Grafting yield: 30.5% — Example 2 - Grafting yield: 11% — Example 2 - Starting polymer SBR No. 6 styrene units: 57.8% — Example 3 - Starting polymer SBR No. 7a styrene units: 57.8% — Example 3 - Starting polymer SBR No. 8 styrene units: 24.3% — Example 3 - Starting polymer SBR No. 10 styrene units: 7.0% — Example 3 - Starting polymer SBR No. 12 styrene units: 5.0% — Example 3 - Reaction temperature: 40° C.

— Example 4 - Reaction temperature: 100° C. — Example 4 - Mercaptoundecanoic acid: 300 meq./kg — Example 4 - Mercaptoundecanoic acid: 350 meq./kg — Example 4 - Mercaptoundecanoic acid: 60 meq./kg — Example 4 - Grafting yield: 71% — Example 4 - Grafting yield: 73% — Example 4 - Grafting yield: 24% — Example 4 - Medium & Water Role: The grafting reaction is performed in organic solution (cyclohexane). Water appears later only in optional coagulation/workup steps using a water/acetone/HCl mixture. - low styrene content: 7.0% styrene

Medium & Water Role - The grafting reaction is performed in organic solution (cyclohexane). Water appears later only in optional coagulation/workup steps using a water/acetone/HCl mixture. - Core reaction medium: Organic solution grafting in cyclohexane; later stripping/coagulation and optional water-containing redissolution/coagulation workup

low styrene content - 7.0% styrene

Relevant Experimental Evidence - Example 1: Antioxidized SBR was grafted with 3-mercaptopropionic acid in cyclohexane and then stripped and dried. - Example 2: Different mercapto acids and antioxidant levels were compared, showing strong dependence of grafting yield on antioxidant content. - Example 3: Grafting was demonstrated on SBRs with widely varying styrene contents, including 7 wt% and 5 wt% examples. - Example 4: Solvent-free grafting at 40° C. and 100° C. was shown, with and without peroxide. - Example 3: The patent shows the grafting strategy can be applied to low-styrene SBR starting materials.

Technical Relevance This patent is relevant because it includes low-styrene SBR starting materials and demonstrates a route to functionalize them without destroying macrostructure. While not a synthesis patent for low-styrene SBR itself, it is important for downstream modification of low-styrene SBR used in tire tread formulations.

Patent 6

- Patent Number: US12037432B2
- Patent Title: Method for preparing modified and conjugated diene-based polymer and modified ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of styrene-butadiene copolymers in n-hexane with n-butyllithium, followed by end-capping, alkoxysilane modification, and a condensation-stabilization step using a modification stabilizer. The polymer is then quenched, treated with antioxidant, and worked up by hot-water solvent removal and roll drying. - Styrene: 270 g — Example 1 - 1,3-butadiene: 710 g — Example

1 - n-hexane: 5,000 g — Example 1 - 2,2-di(2-tetrahydrofuryl)propane: 1.3 g — Example 1 - Temperature: 40° C. — Example 1 - n-butyllithium: 4 mmol — Example 1 - 1,3-butadiene capping: 20 g — Example 1 - Modifier: 4 mmol of a compound of Formula 5a — Example 1 - Ethanol quench: 0.03 parts by weight based on monomers — Example 1 - BHT solution: 5 ml of 0.3 wt % — Example 1 - Condensation reaction time: 15 minutes — Example 1 - Hot water temperature: 100° C. — Example 1 - Roll drying: remove remaining solvents and water — Example 1 - Mooney viscosity: 67 — Experimental Example 1 - Mooney viscosity: 69 — Experimental Example 1 - Mooney viscosity: 73 — Experimental Example 1 - Mooney viscosity: 97 — Experimental Example 1 - Styrene content: 27.0 wt % based on total polymer amount — Experimental Example 1 - Styrene content: 26.8 wt % based on total polymer amount — Experimental Example 1 - Styrene content: 26.9 wt % based on total polymer amount — Experimental Example 1 - Styrene content: 27.1 wt % based on total polymer amount — Comparative/Experimental Example 1 - Vinyl content: 42.9 wt % based on total polymer amount — Experimental Example 1 - Vinyl content: 43.1 wt % based on total polymer amount — Experimental Example 1 - Vinyl content: 42.8 wt % based on total polymer amount — Experimental Example 1 - Vinyl content: 42.9 wt % based on total polymer amount — Comparative/Experimental Example 1 - Mn: 12.6×10^4 g/mol — Experimental Example 1 - Mn: 13.2×10^4 g/mol — Experimental Example 1 - Mn: 13.0×10^4 g/mol — Experimental Example 1 - Mn: 16.9×10^4 g/mol — Comparative/Experimental Example 1 - Mw: 13.6×10^4 g/mol — Experimental Example 1 - Mw: 14.8×10^4 g/mol — Experimental Example 1 - Mw: 15.1×10^4 g/mol — Experimental Example 1 - Mw: 20.4×10^4 g/mol — Comparative/Experimental Example 1 - PDI: 1.08 — Experimental Example 1 - PDI: 1.12 — Experimental Example 1 - PDI: 1.16 — Experimental Example 1 - PDI: 1.21 — Comparative/Experimental Example 1 - Medium & Water Role: The polymerization is carried out in organic solvent (n-hexane). Water appears later only in the hot-water step used to remove solvent before roll drying. - low styrene content: 27.0 wt % based on total polymer amount

Medium & Water Role - The polymerization is carried out in organic solvent (n-hexane). Water appears later only in the hot-water step used to remove solvent before roll drying. - Core reaction medium: Organic solution polymerization in n-hexane

low styrene content - 27.0 wt % based on total polymer amount

Relevant Experimental Evidence - Example 1: A modified SBR was prepared, stabilized by condensation with a modification stabilizer, and hot-water stripped. - Example 2: The same route was repeated with a different stabilizer and similar polymer composition. - Example 3: Another stabilizer variant was used with the same SBR preparation method. - Comparative Example 1: Without the condensation stabilizer, Mooney viscosity was higher after hot-water workup. - Experimental Example 1: Stabilized polymers retained similar styrene/vinyl contents but lower Mooney viscosity than the comparative sample.

Technical Relevance This patent is relevant because it directly discloses solution SBR with about 27 wt% styrene and a process designed to preserve processability after hot-water stripping. It is useful for understanding how low-to-moderate styrene SBR can be modified and stabilized without excessive viscosity growth.

Patent 7

- Patent Number: US12371558B2
- Patent Title: Resin-extended rubber and process for preparing
- Assignee: Bridgestone Americas Tire Operations, Llc
- Jurisdiction: US
- Publication Year: 2025
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of resin-extended rubber by mixing a rubber-solvent-cement containing a conjugated diene monomer-containing rubber with a thermoplastic resin, then removing the non-polar solvent. The examples prepare styrene-butadiene copolymer cements in hexane using n-butyl lithium, then blend with resin before isolation and drying. - Hexane: 4.96 lbs — Preparation of Nanoparticle and Liquid Polymer Blend - 33% styrene in hexane: 0.59 lbs — Preparation of Nanoparticle and Liquid Polymer Blend - 19.3% butadiene in hexane: 3.58 lbs — Preparation of Nanoparticle and Liquid Polymer Blend - Temperature: 120° F. — Preparation of Nanoparticle and Liquid Polymer Blend - Temperature: 117° F. — Preparation of Nanoparticle and Liquid Polymer Blend - n-butyl lithium: 2.3 mL of 1.6 M — Preparation of Nanoparticle and Liquid Polymer Blend - OOPs: 0.76 mL of 1.6 M — Preparation of Nanoparticle and Liquid Polymer Blend - Isopropanol: 0.14 mL — Preparation of Nanoparticle and Liquid Polymer Blend - Additional n-butyl lithium: 2.3 mL — Preparation of Nanoparticle and Liquid Polymer Blend - Styrene blend: 140.2 g — Preparation of Nanoparticle and Liquid Polymer Blend - DVB: 28.5 mL — Preparation of Nanoparticle and Liquid Polymer Blend - Jacket temperature: 180° F. — Preparation of Nanoparticle and Liquid Polymer Blend - Temperature: 90° F. — Preparation of Nanoparticle and Liquid Polymer Blend - Liquid polymer Mw: 73,000 to 80,000 — Preparation of Nanoparticle and Liquid Polymer Blend - Nanoparticle Mw: 83,700 — Preparation of Nanoparticle and Liquid Polymer Blend - Blend composition: 55 wt % liquid polymer and 45 wt % nanoparticles — Preparation of Nanoparticle and Liquid Polymer Blend - SBR replacement: 10 phr — Example 4 - Aromatic oil replacement: about 15 phr — Example 4 - Nanoparticles: 10 phr — Example 4 - Liquid polymer/nanoparticle blend: 22.2 phr — Example 4 - Additional liquid polymer: 2.8 phr — Example 4 - SBR: 100 phr in control — Example 4 - Carbon black: 50 phr — Example 4 - Aromatic oil: 15 phr — Example 4 - Zinc oxide: 3.0 phr — Example 4 - Hydrocarbon resin: 2.0 phr — Example 4 - Santoflex 13: 0.95 phr — Example 4 - Stearic acid: 2.0 phr — Example 4 - Wax: 1.0 phr — Example 4 - Sulfur: ~1.3 phr — Example 4 - Cyclohexylbenzothiazole sulfenamide: 1.4 phr — Example 4 - Diphenylguanidine: 0.20 phr — Example 4 - Mixer speed: 60 rpm — Example 4 - Master batch temperature: 110° C. — Example 4 - Final batch temperature: 75° C. — Example 4 - Molding temperature: 165° C. — Example 4 - Molding time: 15 minutes — Example 4 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - low styrene content: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic hydrocarbon solution polymerization in hexane; no aqueous polymerization medium disclosed

low styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Sample 1: A styrene-butadiene cement was prepared in hexane and later extended with resin before isolation. - Sample 2: Higher initiator loading produced a lower-Mw polymer cement. - Example 4: Nanoparticles and liquid polymer were used to replace SBR and aromatic oil in a rubber compound. - Example 4: The blend was processed in a Brabender mixer and molded at 165° C. - Example 4: The patent demonstrates resin-extended rubber processing rather than a new SBR synthesis route.

Technical Relevance This patent is relevant because it uses solution SBR cements as the base material for resin extension, showing how low-to-moderate styrene SBR can be integrated into downstream formulations. It is less about low-styrene synthesis than about processing and compounding of SBR-containing cements.

Patent 8

- Patent Number: US7829624B2
- Patent Title: One-pot synthesis of nanoparticles and liquid polymer for rubber applications
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-pot solution synthesis in hydrocarbon solvent that first forms a liquid styrene-butadiene polymer, partially terminates it, and then adds styrene and divinylbenzene to generate nanoparticles in situ. The resulting nanoparticle/liquid polymer blend is isolated by isopropanol coagulation, filtration, and drum drying. - Hexane: 4.96 lbs — Preparation of Nanoparticle and Liquid Polymer Blend - 33% styrene: 0.59 lbs — Preparation of Nanoparticle and Liquid Polymer Blend - 19.3% butadiene: 3.58 lbs — Preparation of Nanoparticle and Liquid Polymer Blend - Temperature: 120° F. — Preparation of Nanoparticle and Liquid Polymer Blend - Temperature: 117° F. — Preparation of Nanoparticle and Liquid Polymer Blend - n-butyl lithium: 2.3 mL of 1.6 M — Preparation of Nanoparticle and Liquid Polymer Blend - OOPs: 0.76 mL of 1.6 M — Preparation of Nanoparticle and Liquid Polymer Blend - Isopropanol: 0.14 mL — Preparation of Nanoparticle and Liquid Polymer Blend - Additional n-butyl lithium: 2.3 mL — Preparation of Nanoparticle and Liquid Polymer Blend - Styrene blend: 140.2 g — Preparation of Nanoparticle and Liquid Polymer Blend - DVB: 28.5 mL — Preparation of Nanoparticle and Liquid Polymer Blend - Jacket temperature: 180° F. — Preparation of Nanoparticle and Liquid Polymer Blend - Temperature: 90° F. —

Preparation of Nanoparticle and Liquid Polymer Blend - Liquid polymer Mw: 73,000 to 80,000 — Preparation of Nanoparticle and Liquid Polymer Blend - Nanoparticle Mw: 83,700 — Preparation of Nanoparticle and Liquid Polymer Blend - Blend composition: 55 wt % liquid polymer and 45 wt % nanoparticles — Preparation of Nanoparticle and Liquid Polymer Blend - SBR replacement: 10 phr — Example 4 - Aromatic oil replacement: about 15 phr — Example 4 - Nanoparticles: 10 phr — Example 4 - Liquid polymer/nanoparticle blend: 22.2 phr — Example 4 - Additional liquid polymer: 2.8 phr — Example 4 - SBR: 100 phr in control — Example 4 - Carbon Black: 50 phr — Example 4 - Aromatic Oil: 15 phr — Example 4 - Zinc Oxide: 3.0 phr — Example 4 - Hydrocarbon Resin: 2.0 phr — Example 4 - Santoflex 13: 0.95 phr — Example 4 - Stearic Acid: 2.0 phr — Example 4 - Wax: 1.0 phr — Example 4 - Mixer: 300 g Brabender — Example 4 - Agitation speed: 60 rpm — Example 4 - Master batch temperature: 110° C. — Example 4 - Final batch temperature: 75° C. — Example 4 - Molding temperature: 165° C. — Example 4 - Molding time: 15 minutes — Example 4 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - low styrene content: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic hydrocarbon solution polymerization in hexane

low styrene content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Preparation of Nanoparticle and Liquid Polymer Blend: A one-pot solution process produced a liquid polymer and in situ nanoparticles. - The blend contained 55 wt% liquid polymer and 45 wt% nanoparticles. - Example 4: The blend was used to replace SBR and aromatic oil in a rubber compound. - Example 4: The compound was mixed in a Brabender and molded at 165° C. - The patent demonstrates processing of SBR-containing formulations rather than low-styrene control of SBR composition.

Technical Relevance This patent is relevant because it uses styrene-butadiene copolymer as the liquid-polymer component in a solution process and then deploys the product in rubber formulations. It is only indirectly relevant to low-styrene SBR because the disclosed SBR composition is not the central optimization variable.

Patent 9

- Patent Number: US5902856A
- Patent Title: Rubber composition of two diene rubbers
- Assignee: Nippon Zeon Co., Ltd.
- Jurisdiction: US
- Publication Year: 1999
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses blending a terminal-modified SBR solution with a high-molecular-weight diene rubber solution in solution state, followed by steam stripping, water squeezing, and vacuum drying to obtain a diene rubber. The broader disclosure covers mixed diene-rubber systems designed to balance processability and performance. - Stirring temperature: 60° C. — Example 1 - Stirring time: 30 minutes — Example 1 - Steam stripping: conducted for coagulation and recovery — Example 1 - Vacuum drying temperature: 60° C. — Example 1 - Vacuum drying time: 24 hours — Example 1 - Material rubber: 100 parts — Example 1 - Stearic acid: 1.5 parts — Example 1 - Zinc oxide #1: 3 parts — Example 1 - N-339: 50 parts — Example 1 - Aromatic oil: 5 parts — Example 1 - Sulfur: 1.75 parts — Example 1 - Vulcanization accelerator: 1 part — Example 1 - Terminal-modified polymer A ST/V: 21/63 — Table 1 - Terminal-modified polymer B ST/V: 21/66 — Table 1 - Terminal-modified polymer C ST/V: 21/65 — Table 1 - Terminal-modified polymer D ST/V: 21/65 — Table 1 - Terminal-modified polymer E ST/V: 21/65 — Table 1 - Terminal-modified polymer F ST/V: 21/65 — Table 1 - Terminal-modified polymer G ST/V: 21/64 — Table 1 - High-molecular polymer H ST/V: 21/65 — Table 1 - High-molecular polymer I ST/V: 22/66 — Table 1 - High-molecular polymer J ST/V: 21/64 — Table 1 - M1: 50,000 — Table 1 - M1: 150,000 — Table 1 - M1: 210,000 — Table 1 - M1: 250,000 — Table 1 - M1: 300,000 — Table 1 - M1: 360,000 — Table 1 - M1: 650,000 — Table 1 - M2: 1,200,000 — Table 1 - M2: 900,000 — Table 1 - M2: 600,000 — Table 1 - Molecular weight ratio (M2/M1): 5.5 — Example 3 - Molecular weight ratio (M2/M1): 4.8 — Example 4 - Molecular weight ratio (M2/M1): 2.9 — Example 6 - Molecular weight ratio (M2/M1): 5.0 — Example 7 - Terminal-modified polymer L ST/V: 15/28 — Table 2 - High-molecular polymer M ST/V: 15/29 — Table 2 - Terminal-modified polymer N ST/V: 23/24 — Table 3 - High-molecular polymer O ST/V: 23/35 — Table 3 - Terminal-modified polymer D ST/V: 21/65 — Table 5 - High-molecular polymer H ST/V: 21/65 — Table 5 - Terminal-modified polymer D ST/V: 21/65 — Table 6 - High-molecular polymer H ST/V: 21/65 — Table 6 - Terminal-modified polymer D ST/V: 21/65 — Table 7 - High-molecular polymer H ST/V: 21/65 — Table 7 - Terminal-modified polymer N ST/V: 23/34 — Table 10 - High-molecular polymer U ST/V: 23/35 — Table 10 - Medium & Water Role: The blending is performed in solution state, not in water. Water appears later in steam stripping and in the water squeezing step used during recovery and drying. - low styrene content: 15/28

Medium & Water Role - The blending is performed in solution state, not in water. Water appears later in steam stripping and in the water squeezing step used during recovery and drying. - Core reaction medium: Solution-state blending of rubber cements; downstream steam stripping and water squeezing

low styrene content - 15/28

Relevant Experimental Evidence - Example 1: Solution blending of terminal-modified and high-molecular diene rubbers followed by steam stripping produced a diene rubber. - Example 3: A low-styrene terminal-modified polymer L (15/28 ST/V) was combined with a high-molecular polymer M. - Example 4: Varying the blend ratio of modified and high-molecular polymers improved abrasion resistance and resiliency. - Example 5: Silica-containing compounds showed improved processability and reinforcement with the blended diene rubber. - Example 6: The

blended diene rubber was also used with natural rubber, emulsion SBR, and high-cis polybutadiene.

Technical Relevance This patent is relevant because it explicitly includes low-styrene SBR-like components, including a 15/28 ST/V terminal-modified polymer, and uses solution blending to tune processability. It is not a primary synthesis patent for low-styrene SBR, but it is important for formulation and macrostructure control of SBR-containing rubbers.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	low styrene content	Key Findings
US6858683B2	Polimeri Europa S.P.A.	2005	The polymerization medium is organic solution (cyclohexane/hydrocarbon solvent), not aqueous. Water appears later only in downstream solvent-removal/workup steps, including steam stripping and recovery of damp rubber.	3% by weight	Comparative Example 1: Linear batch SBR was prepared in cyclohexane and isolated by steam stripping and drying.

US12129321B2	Zhuhai Cosmx Battery Co., Ltd.	2024	Water is part of the core polymerization medium as a deionized-water-based emulsion/latex system. Water is also used later for coagulation, washing, and extraction workup of the latex-derived binder.	25 parts styrene	Example 1: A porous cross-linked SBR microsphere binder was prepared in aqueous emulsion and isolated by salt coagulation
US7612144B2	Bridgestone Corporation	2009	No water-related process step disclosed in the extracted evidence.	21.7% styrene	Example No. 2: A styrene-butadiene copolymer was prepared with 2-lithio-2-methyl-1,3-dithiane and showed 21.7% styrene.

EP3030430B1	Compagnie Générale des Etablissements Michelin	2017	The polymerization itself is carried out in organic solvent (methylcyclohexane). Water appears later only in steam stripping used to remove solvent before roll drying.	25 %	Preparation of polymer A: Non-functional SBR control was made in methylcyclohexane and steam stripped.
US7312264B2	Michelin Recherche Et Technique S.A.	2007	The grafting reaction is performed in organic solution (cyclohexane). Water appears later only in optional coagulation/workup steps using a water/acetone/HCl mixture.	7.0% styrene	Example 1: Antioxidized SBR was grafted with 3-mercaptopropionic acid in cyclohexane and then stripped and dried.

US12037432B2	Lg Chem, Ltd.	2024	The polymerization is carried out in organic solvent (n-hexane). Water appears later only in the hot-water step used to remove solvent before roll drying.	27.0 wt % based on total polymer amount	Example 1: A modified SBR was prepared, stabilized by condensation with a modification stabilizer, and hot-water stripping
US12371558B2	Bridgestone Americas Tire Operations, Llc	2025	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Sample 1: A styrene-butadiene cement was prepared in hexane and later extended with resin before isolation.

US7829624B2	Bridgestone Corporation	2010	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Preparation of Nanoparticle and Liquid Polymer Blend: A one-pot solution process produced a liquid polymer and in situ
US5902856A	Nippon Zeon Co., Ltd.	1999	The blending is performed in solution state, not in water. Water appears later in steam stripping and in the water squeezing step used during recovery and drying.	15/28	Example 1: Solution blending of terminal-modified and high-molecular diene rubbers followed by steam stripping produced

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Solution-polymerized SBR patents (US6858683B2, EP3030430B1, US12037432B2, US5902856A) generally use hydrocarbon solvents and lithium-based initiators, while US12129321B2 is the only clearly aqueous emulsion/latex route among the selected set.

- The lowest explicitly disclosed styrene levels in the selected primary patents are 3 wt% feed in US6858683B2 and 7.0 wt% in US7312264B2; several other patents sit in the 20-27 wt% range, including 21.7 wt% (US7612144B2), 25 wt% (EP3030430B1), 25.7 wt% (US7312264B2 starting SBR), and 27.0 wt% (US12037432B2).
- US12129321B2 uses deionized water as the polymerization medium and then salt coagulation, acid treatment, extraction, and washing, whereas the solution patents rely on steam stripping, solvent removal, or coagulation in alcohol.
- Initiator systems vary substantially: n-butyllithium and related organolithiums dominate the solution patents, sulfur-containing lithio initiators are central in US7612144B2, and peroxide/redox emulsion initiation is used in US12129321B2.
- Post-polymerization workup is a major trend across the set: steam stripping and water squeezing in US5902856A and EP3030430B1, hot-water solvent removal in US12037432B2, and coagulation/drying in alcohol for US7612144B2 and US7829624B2.

4. CONCLUSION

Across the selected primary patents, low-styrene SBR is represented mainly through solution-polymerization patents that tightly control monomer feed, conversion, and post-treatment, plus one aqueous emulsion binder patent. The most directly low-styrene disclosures are the 3 wt% styrene feed in US6858683B2 and the 7.0 wt% styrene starting polymer in US7312264B2, while several other patents disclose SBR in the 20-27 wt% styrene range and focus more on processability, branching, or functionalization than on lowering styrene itself. Overall, the landscape emphasizes solvent-based anionic polymerization with lithium initiators, followed by branching, coupling, grafting, or stabilization, and a separate waterborne latex route for microsphere binders.

5. REFERENCES

- US6858683B2 | Process for the preparation of SBR rubbers with an improved processability and ... | Polimeri Europa S.P.A. | US | 2005 | <https://patents.google.com/patent/US6858683B2/en>
- US12129321B2 | Cross-linked SBR microsphere binder and preparation method thereof and lithium- ... | Zhuhai Cosmx Battery Co., Ltd. | US | 2024 | <https://patents.google.com/patent/US12129321B2/en>
- US7612144B2 | Use of sulfur containing initiators for anionic polymerization of monomers | Bridgestone Corporation | US | 2009 | <https://patents.google.com/patent/US7612144B2/en>
- EP3030430B1 | Modified diene elastomer comprising a diene elastomer coupled by an ... | Compagnie Générale des Etablissements Michelin | EP | 2017 | <https://patents.google.com/patent/EP3030430B1/en>
- US7312264B2 | Process for obtaining a grafted elastomer having functional groups along the ... | Michelin Recherche Et Technique S.A. | US | 2007 |

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- US12037432B2 | Method for preparing modified and conjugated diene-based polymer and modified ... | Lg Chem, Ltd. | US | 2024 | <https://patents.google.com/patent/US12037432B2/en>
- US12371558B2 | Resin-extended rubber and process for preparing | Bridgestone Americas Tire Operations, Llc | US | 2025 | <https://patents.google.com/patent/US12371558B2/en>
- US7829624B2 | One-pot synthesis of nanoparticles and liquid polymer for rubber applications | Bridgestone Corporation | US | 2010 | <https://patents.google.com/patent/US7829624B2/en>
- US5902856A | Rubber composition of two diene rubbers | Nippon Zeon Co., Ltd. | US | 1999 | <https://patents.google.com/patent/US5902856A/en>